STUDY OF IMMUNOGENIC CELL DEATH MARKERS IN LUNG ADENOCARCINOMA CELL LINES TREATED WITH THERAPEUTIC AGENTS

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ABSTRACT

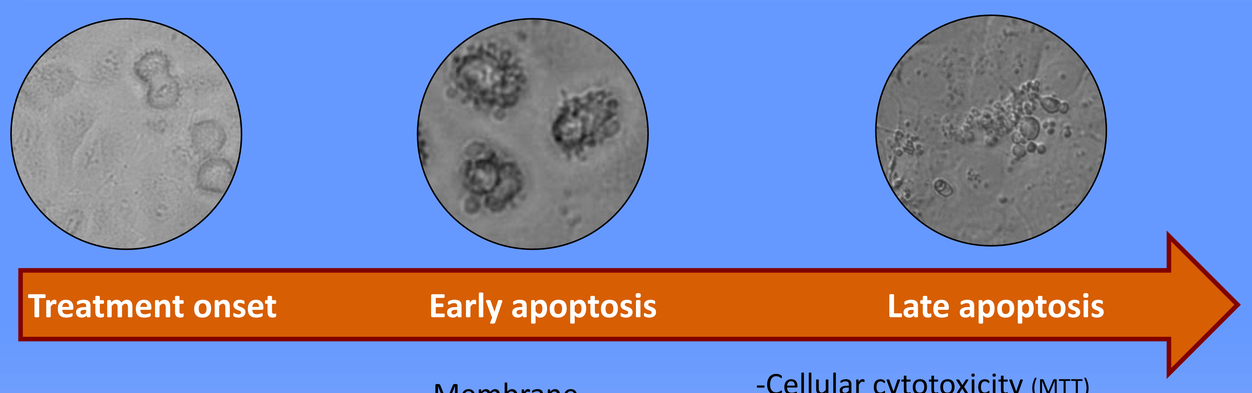
Lung cancer is the leading cause of cancer-related mortalities worldwide and cisplatin (CDDP) is the conventional treatment. Recently in patients, *EGFR* mutation-positive lung adenocarcinoma has been detected. In these group of patients, Tyrosine Kinase Inhibitors (TKIs) treatment is indicated. Erlotinib, a TKI, induces in sensitive cell lines cell cycle arrest and apoptosis. These biological events are also induced by CDDP.

Immunogenic cell death (ICD) is a type of cell death induced by some cytotoxic drugs in which intracellular molecules are emitted for activating the antitumor immune response. These molecules are collectively named as Damage Associated Molecular Patterns (DAMPs). ICD requires the spatial-temporal expression of three essential DAMPs: membrane exposed calreticulin and release of ATP and HMGB1. Currently, no reports exist whether Erlotinib and CDDP induces ICD in EGFR-mutated and wild type lung cancer cell lines, respectively.

OBJECTIVE

Analyze in the A549, HCC4006 and HCC827 lung adenocarcinoma cell lines, whether CDDP or Erlotinib induce cell death associated with ICD markers expression

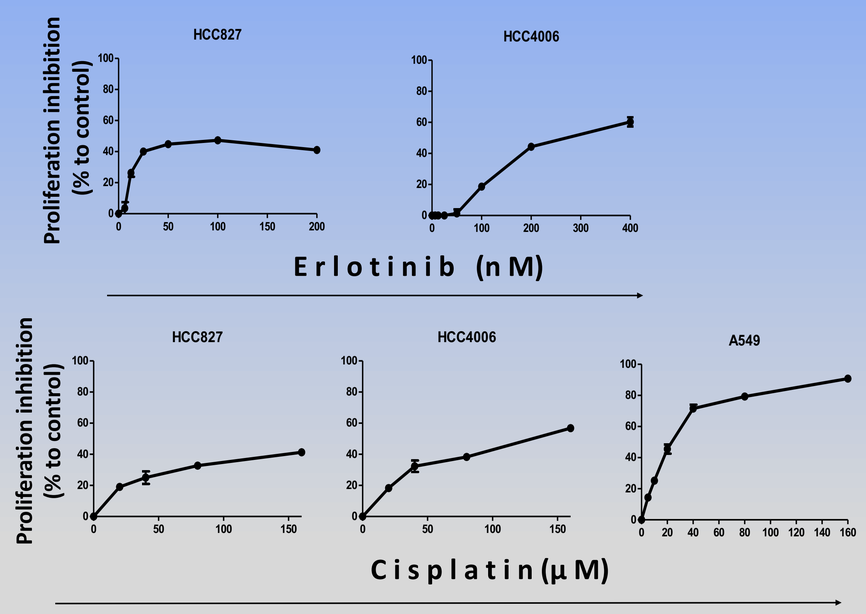
METHODOLOGY



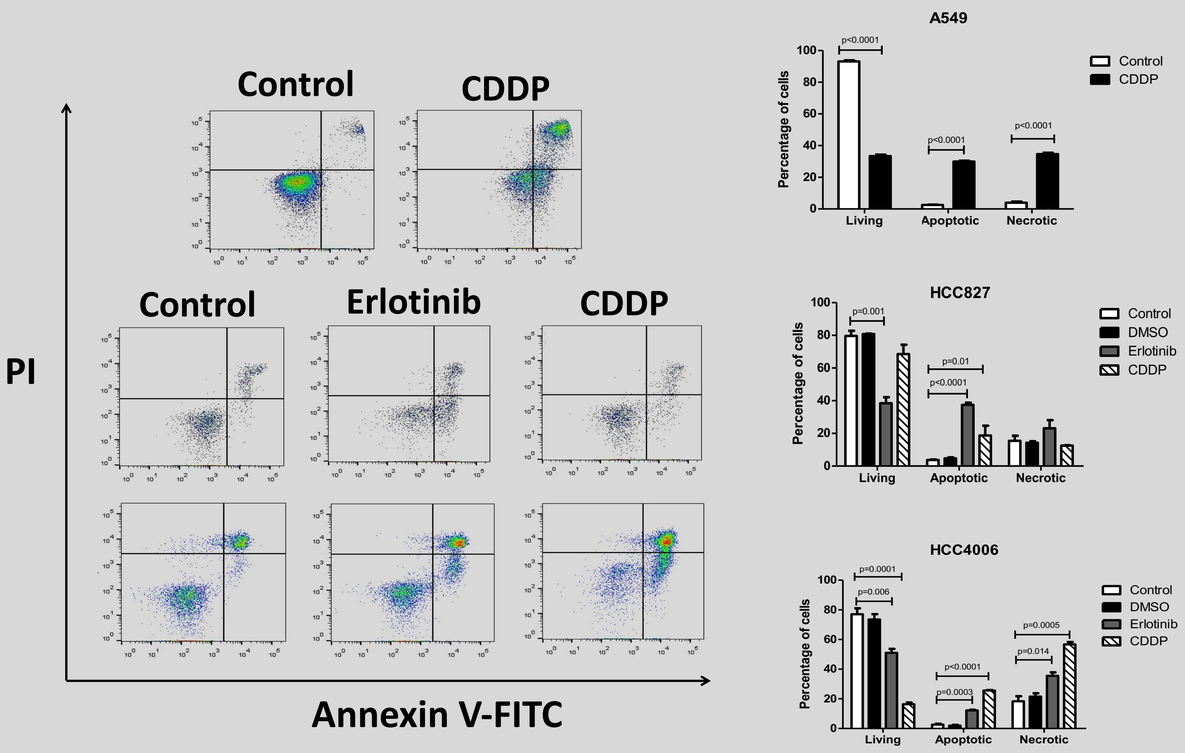
-Membrane calreticulin (Flow cytometry and Immunofluorescence)

-Cellular cytotoxicity (MTT)
-Cell death (Flow cytometry)
-Intracellular ATP & ADP (Luminiscence)
-HMGB1 cell distribution (Immunofluorescence)
-HMGB1 release (ELISA)

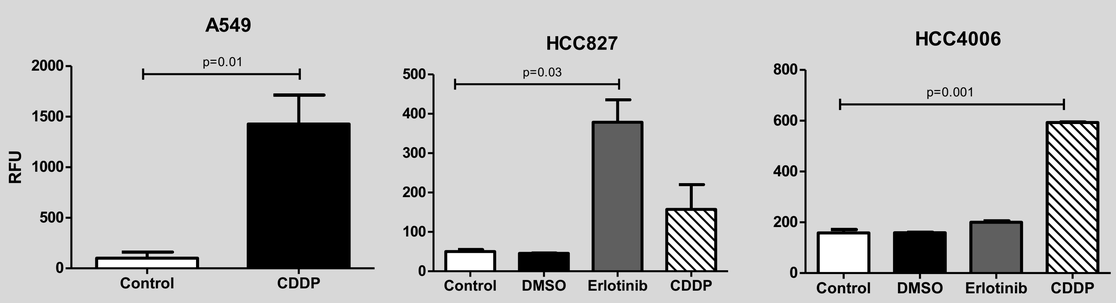
RESULTS



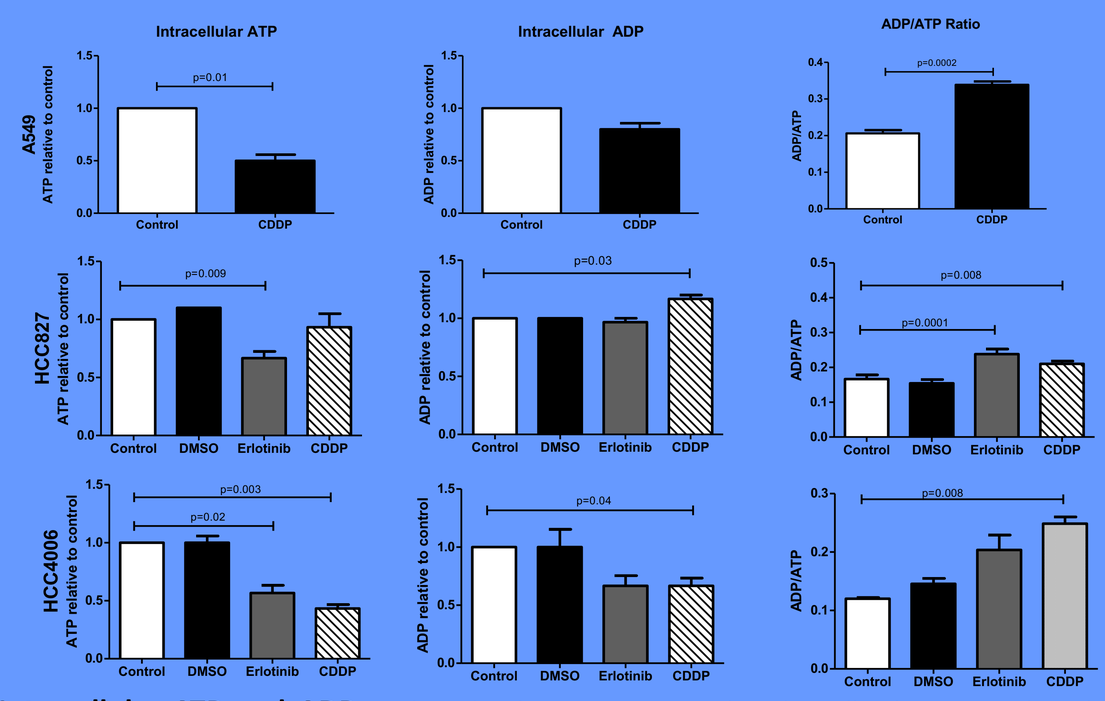
Dose-response curves of Erlotinib (48 h) and CDDP (24 h) treatment. Each experiment was done in triplicate and data were obtained from three independent experiments. Mean ± SEM are indicated



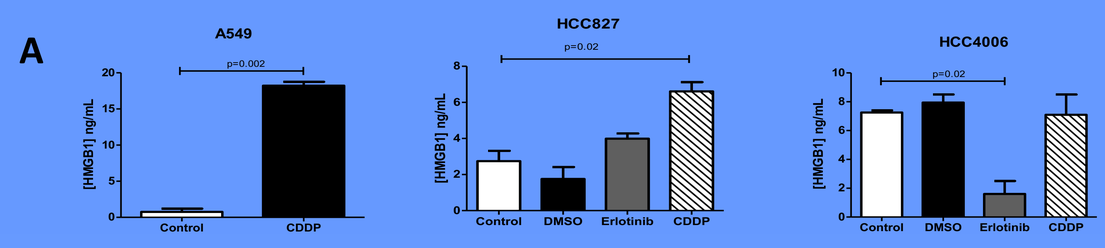
Apoptosis induction after drug treatment in lung adenocarcinoma cell lines. Apoptosis was detected after CDDP (16h) or Erlotinib (40h) treatment. Each experiment was done in triplicate and data were obtained from three independent experiments. Mean ± SEM are indicated.

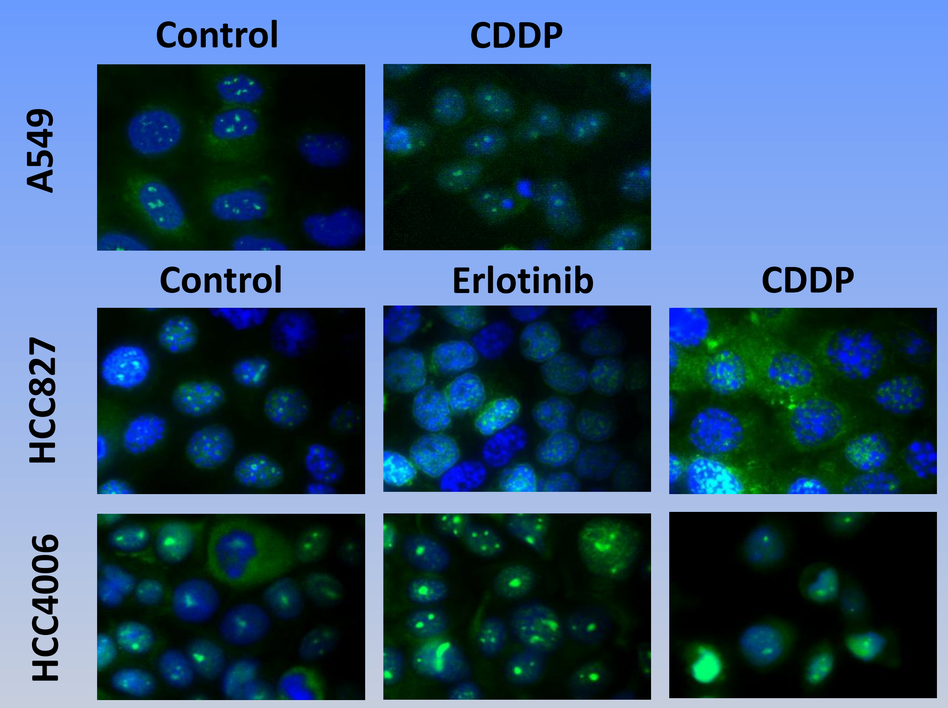


CDDP and Erlotinib increase caspase-3 activity in lung adenocarcinoma cell lines. Caspase-3 activity was studied by fluorometry. Each experiment was done in triplicate and data were obtained from three independent experiments. Mean ± SEM are indicated.

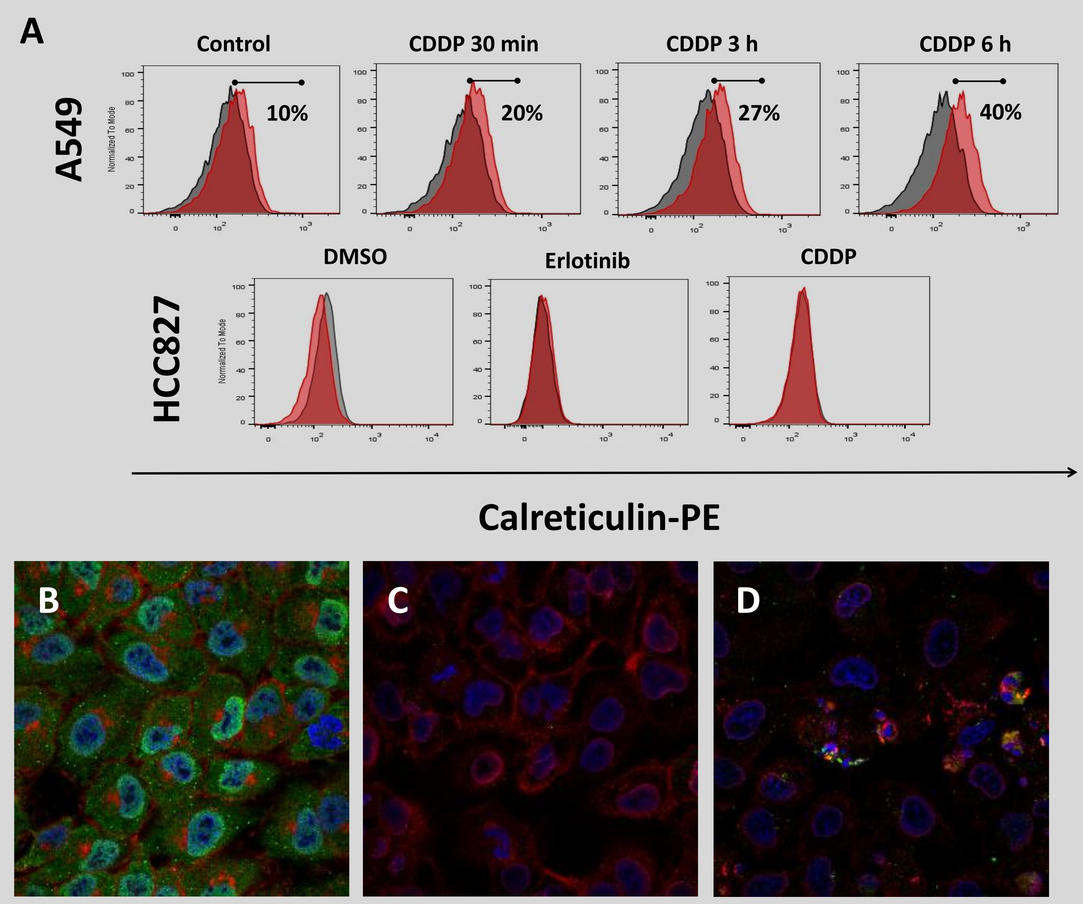


Intracellular ATP and ADP. Both molecules were measured in the cell cultures at 15h and 40h of CDDP and Erlotinib treatment, respectively. The ADP/ATP ratio is indicated. Mean ± SD are shown.





HMGB1 in the lung adenocarcinoma cell lines. HMGB1 concentrations in **supernatants** from cell cultures at the end of treatments (A). Mean ± SD are shown. In **residual cells,** intracellular distribution of HMGB1 at the end of treatments (B). Original magnification 100 X. HMGB1 is in green and nuclear staining in blue (DAPI).



Calreticulin exposition in lung adenocarcinoma cell lines. Kinetics of calreticulin membrane-exposition in A549 and HCC827 (24h of treatment) cell lines (A). In A549, intracellular calreticulin in permeabilized non-treated cells (B); no membrane exposition in non-treated cells (C) and membrane exposition on CDDP-treated and non-permeabilized cells after 6 h (D). Calreticulin in green, membrane staining in red (WGA-Alexa633) and nuclear staining in blue (DAPI). Original magnification 60 X.

CONCLUSIONS

In A549 cell line, CDDP induced apoptosis associated with expression of ICD markers. This phenomenon could lead to the activation of an antitumor immune response in CDDP-treated patients.

In EGFR-mutated cell lines, Erlotinib-induced cell death is not mediated by ICD. Further studies are required to evaluate the potential of other TKI's in inducing ICD.